Sun-Tracking Camera Design Package

Prototype ESP32-CAM + four-LDR sun sensor + dual-servo pan/tilt concept powered from a 5 V adapter.

Goal

Design a prototype sun-tracking camera using an ESP32-CAM module, a two-axis pan/tilt mechanism, and four light sensors so the camera can automatically steer toward the brightest direction while providing live image capture over Wi-Fi.

Assumptions

• User selected a prototype dev-board style implementation.
• Power source is a regulated 5 V DC adapter.
• This is a conceptual/prototyping package, not a production-ready outdoor engineering release.
• Exact supplier stock and pricing were not verified live; links are representative product pages.
• Chosen controller is AI-Thinker ESP32-CAM, which commonly requires a strong 5 V supply because camera + Wi-Fi + servo transients can cause brownouts.
• Two SG90 micro servos are acceptable for lightweight indoor demonstration mechanics.

Basic circuit description

The design has four main blocks:

1. **5 V power distribution**: A 5 V / >=2 A adapter feeds the ESP32-CAM 5 V pin and both SG90 servos in parallel. A 470 uF bulk capacitor and 100 nF bypass capacitor are placed across the 5 V rail near the servo/header area to suppress supply dips and high-frequency noise.

2. **Camera/control block**: An AI-Thinker ESP32-CAM module serves as the central processor and image sensor. It streams video over Wi-Fi and reads four analog voltages from light-dependent resistor divider networks. It also generates PWM outputs for the two servo control lines.

3. **Sun direction sensing block**: Four LDRs are arranged physically in a quadrant around a small vertical shade cross. Each LDR is paired with a 10 kOhm resistor to form a voltage divider. Differential comparison of left-vs-right and top-vs-bottom light levels estimates where the sun is relative to the current pointing direction.

4. **Actuation block**: Two SG90 servos move the pan and tilt axes. The ESP32-CAM drives one servo for azimuth and one for elevation. Control firmware moves only in small steps when light imbalance exceeds a threshold, reducing jitter.

Recommended signal assignments
Because ESP32-CAM pin availability is constrained by the camera interface, use a build that exposes ADC-capable pins for the four LDR dividers and two PWM-capable GPIOs for servos. A practical prototype assumption is:
• Pan servo signal: GPIO14
• Tilt servo signal: GPIO15
• LDR dividers: ADC-capable available inputs on the chosen ESP32-CAM breakout/carrier implementation

If the specific carrier board does not expose enough ADC pins, add a small external ADC (for example ADS1115 over I2C) or use an ESP32 camera board variant with more broken-out pins.

Main components

• U1: ESP32-CAM AI-Thinker module with OV2640 camera
• M1, M2: SG90 5 V micro servos
• R1-R4: LDR sensors in four-quadrant layout
• R5-R8: 10 kOhm divider resistors

- C1: 470 uF electrolytic bulk capacitor
- C2: 100 nF ceramic bypass capacitor
- J1: 5 V DC input connector
- J2, J3: servo headers
- J4: FTDI programming header
- MECH1: pan-tilt bracket

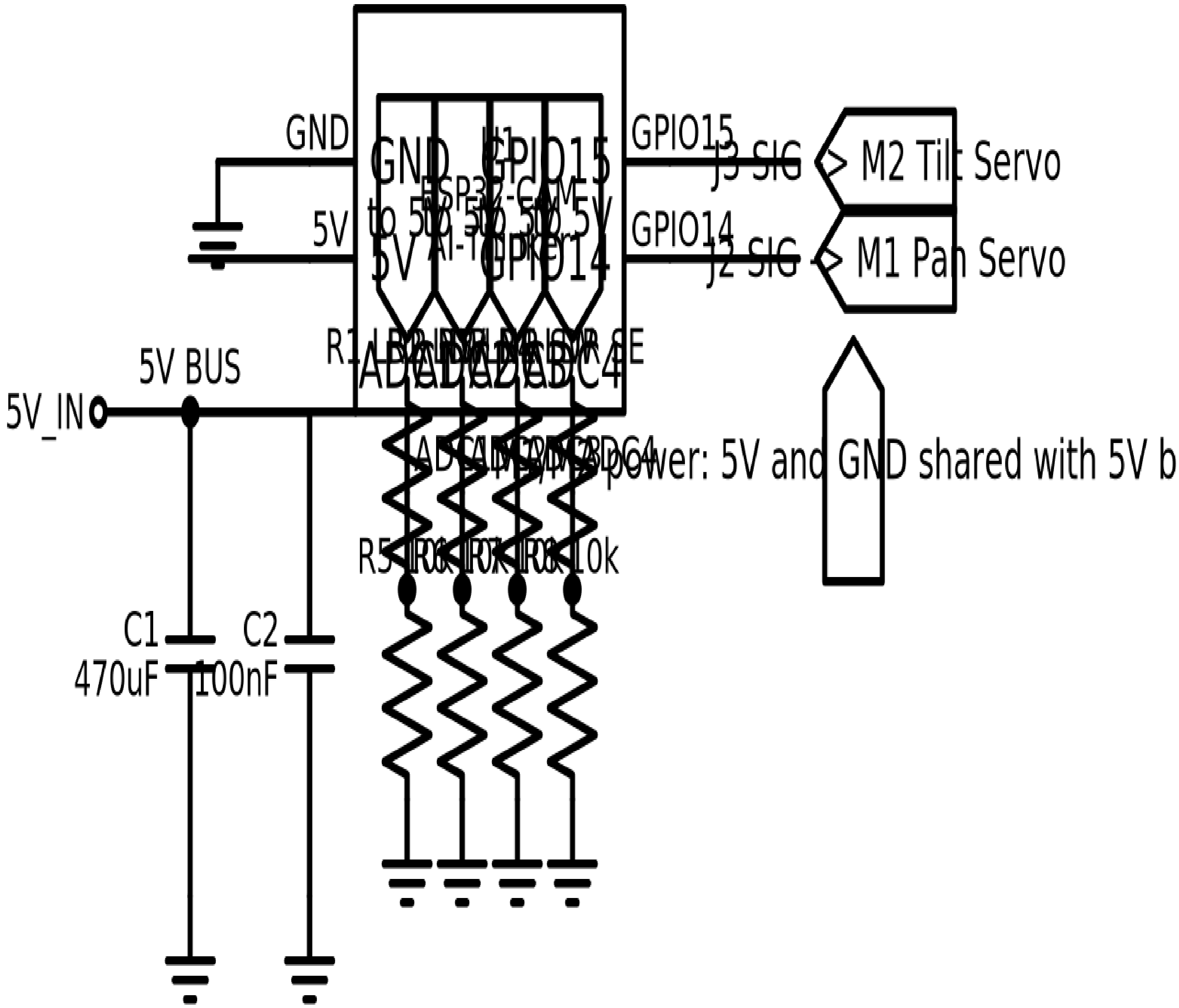
Important unknowns / limitations

- A bare ESP32-CAM board has limited accessible ADC pins; the final wiring may depend on the exact carrier/backplane used.
- SG90 servos are acceptable for a lightweight camera head, but wind loading or outdoor use may require metal-geared servos and a separate servo supply rail.
- LDR-based solar pointing is simple but not high precision. For astronomy-grade solar tracking, use ephemeris/GPS/RTC or a calibrated sun sensor.
- Outdoor deployment would require enclosure, UV/weather protection, cable strain relief, thermal design, and likely different mechanics.

Firmware concept

- Read four sensor voltages repeatedly.
- Compute horizontal error = (right pair average) - (left pair average).
- Compute vertical error = (top pair average) - (bottom pair average).
- If horizontal error exceeds threshold, increment/decrement pan servo.
- If vertical error exceeds threshold, increment/decrement tilt servo.
- Stream images over Wi-Fi for observation and tuning.
- Optionally add deadband and time averaging to prevent oscillation.

Schematic / Wiring Overview



Note: ADC pin labels in the diagram are conceptual placeholders for accessible analog inputs on the chosen carrier/backplane. A bare ESP32-CAM may need an external ADC or a more capable breakout.

Bill of Materials

Item	RefDes	Qty	Description	Recommended MPN	Supplier	Purchase URL
1	U1	1	ESP32-CAM AI-Thinker module with OV2640	ESP32-CAM (AI-Thinker)	Ai-Thinker / distributors	link
2	M1	1	SG90 micro servo for pan axis	TowerPro SG90 or equivalent	Adafruit	link
3	M2	1	SG90 micro servo for tilt axis	TowerPro SG90 or equivalent	Adafruit	link
4	PS1	1	5 V regulated DC supply	Mean Well GST10A05-P1J	DigiKey	link
5	C1	1	Bulk electrolytic capacitor	EEU-FR1A471	DigiKey	link
6	C2	1	Ceramic decoupling capacitor	C315C104M5U5TA	DigiKey	link
7	R1,R2,R3,R4	4	Photoresistor (LDR) for sun-direction sensing	GL5528	Adafruit	link
8	R5,R6,R7,R8	4	Fixed resistor for LDR dividers	CFR-25JB-52-10K	Mouser	link
9	J1	1	2.1 mm DC barrel jack breakout or terminal input	PRT-00119 or equivalent	SparkFun	link
10	J2,J3	2	3-pin servo headers	Generic 3-pin header	DigiKey	link
11	J4	1	FTDI programming header	Generic 1x6 header	DigiKey	link
12	MECH1	1	Pan-tilt bracket set	Generic SG90 pan-tilt kit	Adafruit	link

Disclaimer: This package is suitable for prototyping and concept communication. Mechanical loading, thermal behavior, outdoor survivability, EMC, and long-term power integrity are not validated here.